

KANISHKA THAKUR

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SUMMARY

Third-year Computer Science Engineering (Data Science) student with a solid foundation in data structures, algorithms, and object-oriented programming built through coursework and hands-on practice. Currently a Software Engineering Intern contributing to backend development, REST API implementation, and RAG pipeline work on a production-adjacent Node.js/TypeScript system. Comfortable with Java-based OOP development, Git-based version control, and Agile collaboration; quick to learn new technologies and pick up unfamiliar codebases.

Core Competencies: Software Engineering, Software Development, Java, Object-Oriented Programming, Data Structures & Algorithms, Problem Solving, REST APIs, Backend Development, Debugging, Testing, Version Control, Team Collaboration, Code Reviews, Engineering Best Practices

EXPERIENCE

Software Engineering Intern — Dotnitrn Pvt. Ltd.

June 2026 – Present

- Contributed to migrating sections of the RAG pipeline codebase from JavaScript to TypeScript to improve type safety and long-term maintainability.
- Integrated and debugged the LiteParse document-parsing module within the ingestion pipeline under senior engineer guidance.
- Developed and tested REST API endpoints in Express.js to support document ingestion and retrieval workflows, using Postman for API testing.
- Worked with the Qdrant vector database to store and query document embeddings as part of the RAG pipeline implementation.
- Wrote engineering documentation and collaborated with backend teammates using Git and GitHub within an Agile development workflow.

Social Impact Intern — Womenite (NGO)

June 2025 – July 2025

- Planned and executed offline community-awareness campaigns and designed online campaigns across digital platforms to support Womenite's outreach initiatives.
- Utilized social media channels to reach target audiences and collaborated with cross-functional teams, contributing to the organization's engagement efforts.

PROJECTS

Modern RAG Pipeline | TypeScript, Node.js, Express.js, Qdrant, LiteParse

- Built a self-directed retrieval-augmented generation prototype covering document ingestion, parsing, and retrieval, using LiteParse for document parsing and Qdrant for vector storage.
- Implemented REST API endpoints in Express.js to expose ingestion and retrieval functionality for testing.

Buy & Sell Management System | Java, Java Swing, OOP, Multithreading, File Handling

- Developed a desktop buy-and-sell management application in Java with a Swing-based GUI, implementing buyer and seller roles, registration, login, and product management.
- Applied object-oriented design and multithreading to handle concurrent buying and selling operations, using file handling for data persistence.
- Published the project on GitHub with a structured README, screenshots, and organized source code documenting the development process.

Notes Manager Web Application | HTML, CSS, JavaScript

- Built a fully functional CRUD note-taking application using LocalStorage-based persistence with JSON serialization.
- Implemented a responsive layout with CSS Flexbox and media queries for consistent usability across devices.

Inflation Forecasting Dashboard | Python, Pandas, Prophet, ARIMA, SARIMA, Streamlit, Plotly

- Preprocessed time-series data and tested stationarity across three forecasting models (ARIMA, SARIMA, Prophet).
- Built an interactive Streamlit dashboard with real-time model switching to support faster model comparison.

TECHNICAL SKILLS

Languages: Java, Python, C (Intermediate), SQL, JavaScript (Basic), TypeScript (working knowledge)

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Debugging, Testing

Web & Backend: HTML, CSS, JavaScript, Node.js, Express.js, REST APIs

Tools & Practices: Git, GitHub, VS Code, Postman, Docker (basic), Agile, Code Reviews

Data & Analytics: Pandas, Streamlit, Plotly, Time-Series Forecasting (ARIMA, SARIMA, Prophet)

Exposure To: LiteParse, Qdrant, Retrieval-Augmented Generation (RAG)

EDUCATION

University of Petroleum and Energy Studies (UPES), Dehradun

Expected 2028

Bachelor of Technology (B.Tech) – Computer Science Engineering (Data Science), Third Year

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Computer Networks, Operating Systems, Discrete Mathematics

HACKATHONS & ACTIVITIES

SATHACK'25 — Thapar Institute of Engineering & Technology

November 2025

- Collaborated in a 5-person team to build a project prototype within a 36-hour hackathon window, dividing frontend and backend responsibilities to ship a functional demo under time pressure.